

PYTHON FUNDAMENTALS FOR DATA ANALYTICS

Week 1 Assignment Report

Submitted By

Name: T. Vaishnavi

Branch: Computer Science and Engineering (CSE)

College: Sri Chandrasekharendra Saraswathi Viswa Mahavidyalaya (SCSVMV)

Course Details

Course Name: Python Fundamentals for Data Analytics

Assignment: Week 1 Assignment

Duration: 1 Week (7 Days)

Total Marks: 100

Introduction

Python is one of the most popular and beginner-friendly programming languages used in Data Analytics, Artificial Intelligence, Machine Learning, Web Development, and Automation. Its simple syntax and powerful features make it easy to write, understand, and maintain programs. Python also provides a wide range of built-in functions and libraries that simplify data analysis and problem-solving.

This assignment focuses on the fundamental concepts of Python programming, including variables, data types, operators, conditional statements, loops, functions, strings, collections, file handling, and the development of a mini project. By completing these tasks, students gain practical programming experience and build a strong foundation for Data Analytics and future software development.

Objective

The objective of this assignment is to understand the fundamentals of Python programming by implementing basic programming concepts and applying them to develop a simple Student Record Management System. This assignment helps strengthen problem-solving skills and prepares students for advanced topics in Data Analytics.

Task 1 – Python Basics

Aim

To understand Python input and output statements.

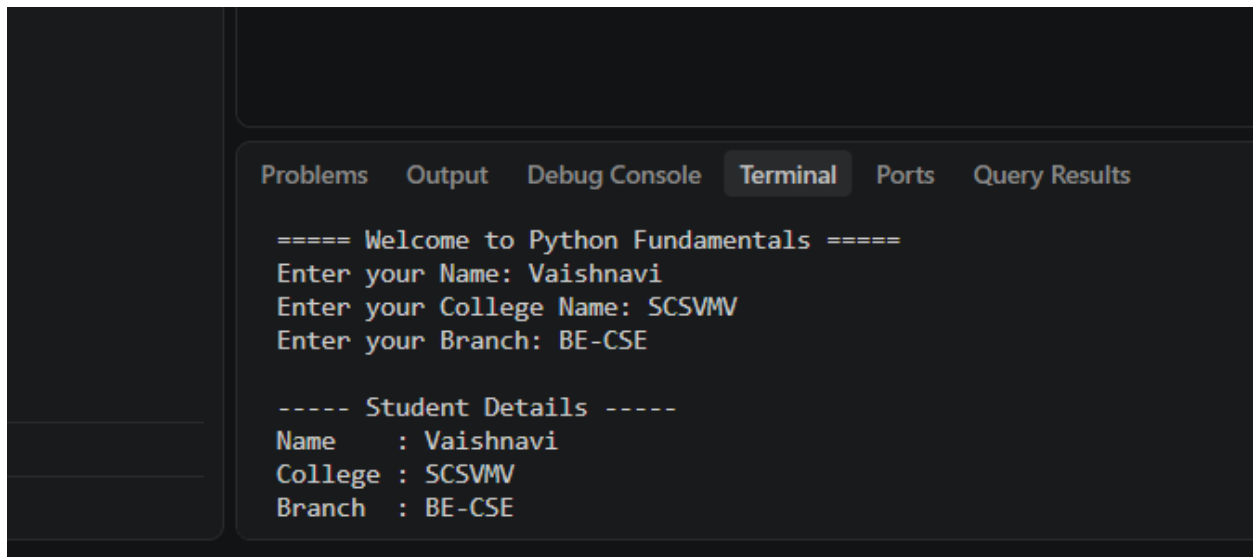
Explanation

This program prints a welcome message, accepts the user's name, college name, and branch as input, and displays the entered information in a formatted manner.

Python Code

```
print("Welcome to Python Programming!")  
  
name = input("Enter your Name: ")  
college = input("Enter your College Name: ")  
branch = input("Enter your Branch: ")  
  
print("\n----- Student Details -----")  
  
print("Name   :", name)  
print("College :", college)  
print("Branch  :", branch)
```

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'Problems', 'Output', 'Debug Console', 'Terminal' (which is selected), 'Ports', and 'Query Results'. The terminal displays the following text:

```
===== Welcome to Python Fundamentals =====  
Enter your Name: Vaishnavi  
Enter your College Name: SCSVMV  
Enter your Branch: BE-CSE  
  
----- Student Details -----  
Name      : Vaishnavi  
College   : SCSVMV  
Branch    : BE-CSE
```

TASK 2 – VARIABLES & DATA TYPES

Aim

To understand different data types in Python.

Explanation

This program creates variables of Integer, Float, String, and Boolean data types. It displays each variable along with its data type using the **type()** function.

Python Code

```
age = 20  
height = 5.7  
name = "Vaishnavi"  
is_student = True  
  
print(age, type(age))  
print(height, type(height))  
print(name, type(name))  
print(is_student, type(is_student))
```

Output:

```
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "TASK-02 VARIABLES & DATATYPES.py"
>> 20 <class 'int'>
>> 5.7 <class 'float'>
>> Vaishnavi <class 'str'>
>> True <class 'bool'>
```

TASK 3 – OPERATORS

Aim

To perform arithmetic operations using Python operators.

Explanation

This calculator program accepts two numbers from the user and performs addition, subtraction, multiplication, division, and modulus operations.

Python Code

```
num1 = float(input("Enter First Number: "))
num2 = float(input("Enter Second Number: "))

print("Addition =", num1 + num2)
print("Subtraction =", num1 - num2)
print("Multiplication =", num1 * num2)
print("Division =", num1 / num2)
print("Modulus =", num1 % num2)
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports  Query Results

PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "TASK-03 OPERATIONS.py"
Enter First Number: 6
Enter Second Number: 6
Addition = 12.0
Subtraction = 0.0
Multiplication = 36.0
Division = 1.0
Modulus = 0.0
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> |
```

TASK 4 – CONDITIONAL STATEMENTS

Aim

To understand the use of **if**, **elif**, and **else** statements for decision-making in Python.

Explanation

This program accepts the student's marks as input and displays the corresponding grade using conditional statements. It checks the marks against different conditions and prints the appropriate grade.

Python Code

```
marks = int(input("Enter Marks: "))
```

```
if marks >= 90:
    print("Grade: A")
elif marks >= 75:
    print("Grade: B")
elif marks >= 60:
    print("Grade: C")
else:
    print("Fail")
```

Output:

```
MODULUS = 0.0  
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "TASK-04 CONDITIONAL STATEMENTS.py"  
Enter Marks: 98  
Grade: A  
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> |
```

TASK 5 – LOOPS

Aim

To understand the use of **for** and **while** loops in Python.

Explanation

This task demonstrates the use of looping statements by:

- Printing numbers from 1 to 20 using a **for** loop.
- Printing the multiplication table of any number using a **for** loop.
- Printing even numbers from 1 to 50 using a **while** loop.

Program 1 – Print Numbers from 1 to 20

Python Code

```
for i in range(1, 21):  
    print(i)
```

Output:

```
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task5_1_numbers_1_to_20.py"
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

Program 2 – Multiplication Table

Python Code

```
num = int(input("Enter Number: "))

for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```


Output:

```
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task5_2_multiplication_table.py"
Enter Number: 8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

Program 3 – Even Numbers from 1 to 50

Python Code

```
i = 2

while i <= 50:
    print(i)
    i += 2
```

Output

```
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task5_3_even_numbers_while.py"
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

TASK 6 – FUNCTIONS

Aim

To understand the creation and use of user-defined functions in Python.

Explanation

This program demonstrates two user-defined functions:

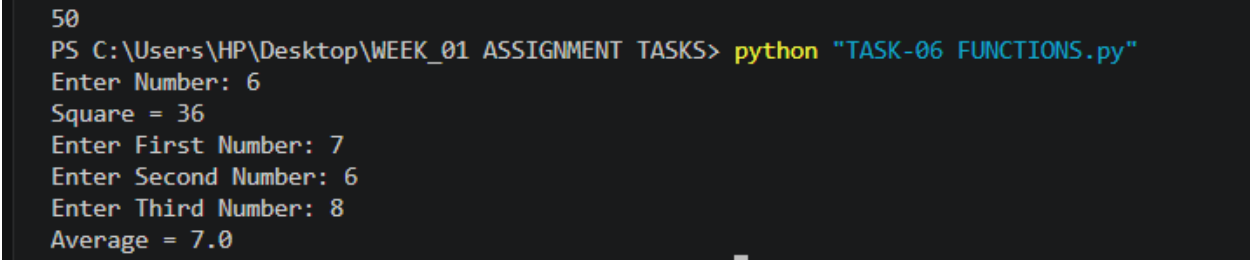
- A function to calculate the square of a number.
- A function to calculate the average of three numbers.

The program accepts input from the user, calls both functions, and displays the results.

Python Code

```
def square(num):  
    return num * num  
  
def average(a, b, c):  
    return (a + b + c) / 3  
  
number = int(input("Enter a Number: "))  
print("Square =", square(number))  
  
x = int(input("Enter First Number: "))  
y = int(input("Enter Second Number: "))  
z = int(input("Enter Third Number: "))  
  
print("Average =", average(x, y, z))
```

Output



```
50  
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "TASK-06 FUNCTIONS.py"  
Enter Number: 6  
Square = 36  
Enter First Number: 7  
Enter Second Number: 6  
Enter Third Number: 8  
Average = 7.0
```

TASK 7 – STRINGS & COLLECTIONS

Aim

To understand string manipulation and the use of different collection data types in Python such as List, Tuple, Dictionary, and Set.

Explanation

This task demonstrates various string methods and collection operations. It includes converting strings to uppercase and lowercase, replacing characters, finding characters, performing list operations, creating tuples, storing data in dictionaries, and performing set operations.

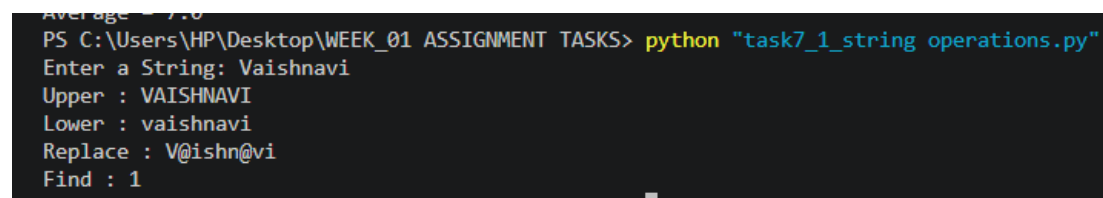
Program 1 – String Operations

Python Code

```
text = input("Enter a String: ")

print("Upper :", text.upper())
print("Lower :", text.lower())
print("Replace :", text.replace("a", "@"))
print("Find :", text.find("a"))
```

Sample Output



```
Average = 7.0
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task7_1_string operations.py"
Enter a String: Vaishnavi
Upper : VAISHNAVI
Lower : vaishnavi
Replace : V@ishn@vi
Find : 1
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS>
```

Program 2 – List Operations

Python Code

```
fruits = ["Apple", "Banana", "Mango"]

fruits.append("Orange")
print("After Append :", fruits)

fruits.remove("Banana")
```

```
print("After Remove :", fruits)
```

```
fruits.sort()
```

```
print("After Sort :", fruits)
```

Output

```
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task7_2_List operations.py"
['Apple', 'Mango', 'Orange']
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

Program 3 – Tuple

Python Code

```
numbers=(10,20,30,40,50)
```

```
print("Tuple:", numbers)
```

```
print("First Element:",numbers[0])
```

```
print("Last Element:", numbers[-1])
```

Output:

```
['Apple', 'Mango', 'Orange']
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task7_3_Tuple operations.py"
Tuple: (10, 20, 30, 40, 50)
First Element: 10
Last Element: 50
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

Program 4 – Dictionary

Python Code

```
student = {
    "Name": "Vaishnavi",
    "Age": 20,
    "Branch": "CSE"
```

```
}
```

```
print(student)
```

Output

```
Last Element: 50
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task7_4_Dictionary operations.py"
{'Name': 'Vaishnavi', 'College': 'SCSVMV', 'Branch': 'CSE', 'Age': 20}
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

Program 5 – Set Operations

Python Code

```
colors = {"Red", "Green", "Blue"}
```

```
colors.add("Yellow") # Add a new element colors.remove("Blue") #
Remove an element
```

```
print(colors)
```

Output

```
{'Name': 'Vaishnavi', 'College': 'SCSVMV', 'Branch': 'CSE', 'Age': 20}
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "task7_5_Sets operations.py"
{'Green', 'Yellow', 'Red'}
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> █
```

TASK 8 – BASIC FILE HANDLING

Aim

To understand file creation, writing, reading, and closing files using Python.

Explanation

This program creates a text file named **intro.txt**, writes a short introduction into it, reads the contents of the file, and displays them on the screen.

Python Code

Creating and writing to a file

```
file = open("intro.txt", "w")

file.write("Hello!\n") file.write("My name is Vaishnavi.\n")
file.write("I am studying Computer Science Engineering.\n")
file.write("I am learning Python for Data Analytics.")

file.close()
```

Reading the file

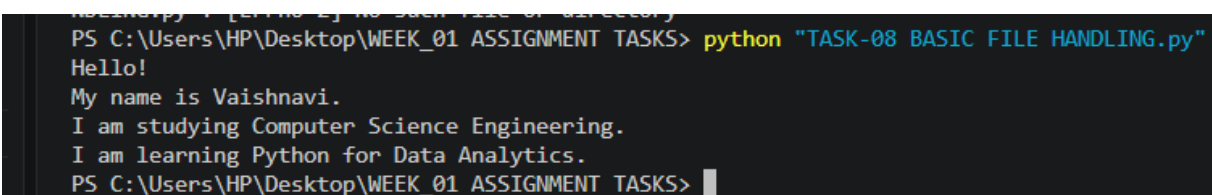
```
file = open("intro.txt", "r")

content = file.read()

print(content)

file.close()
```

Output



```
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "TASK-08 BASIC FILE HANDLING.py"
Hello!
My name is Vaishnavi.
I am studying Computer Science Engineering.
I am learning Python for Data Analytics.
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> |
```

TASK 9 – MINI PYTHON PROJECT

Student Record Management System

Aim

To develop a menu-driven Student Record Management System using Python.

Explanation

This mini project demonstrates the practical application of Python fundamentals. It allows users to add student records, display all records, search for a student by name, delete student records, and exit the application. The program uses lists and dictionaries for storing student information.

Python code:

```
students = []

while True:

    print("\n===== Student Record Management =====")
    print("1. Add Student")
    print("2. Display Students")
    print("3. Search Student")
    print("4. Delete Student")
    print("5. Exit")

    choice = input("Enter Choice: ")
```



```
if choice == "1":
```

```
    name = input("Enter Name: ")
```

```
    age = input("Enter Age: ")
```

```
    branch = input("Enter Branch: ")
```

```
    student = {
```

```
        "Name": name,
```

```
        "Age": age,
```

```
        "Branch": branch
```

```
    }
```

```
    students.append(student)
```

```
    print("Student Added Successfully.")
```

```
elif choice == "2":
```

```
    if len(students) == 0:
```

```
        print("No Student Records Found.")
```

```
    else:
```

```
        for s in students:
```

```
            print(s)
```

```
elif choice == "3":
```

```
search = input("Enter Name to Search: ")
```

```
found = False
```

```
for s in students:
```

```
    if s["Name"].lower() == search.lower():
```

```
        print(s)
```

```
        found = True
```

```
if not found:
```

```
    print("Student Not Found.")
```

```
elif choice == "4":
```

```
delete = input("Enter Name to Delete: ")
```

```
found = False
```

```
for s in students:
```

```
    if s["Name"].lower() == delete.lower():
```

```
        students.remove(s)
```

```
        found = True
```

```
        print("Student Deleted Successfully.")
```

```
        break
```

```
if not found:
```

```
    print("Student Not Found.")
```

```
elif choice == "5":
```

```
    print("Thank You!")
```

```
    break
```

```
else:
```

```
    print("Invalid Choice.")
```

OUTPUT:

```
5. Exit
Enter Choice: 5
Thank You!
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> python "Mini Python Project.py"

===== Student Record Management =====
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter Choice: 1
Enter Name: Vaishnavi
Enter Age: 20
Enter Branch: BE-CSE
Student Added Successfully.

===== Student Record Management =====
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter Choice: 1
Enter Name: Dhanushka
Enter Age: 21
Enter Branch: ECE
Student Added Successfully.

===== Student Record Management =====
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter Choice: 2
{'Name': 'Vaishnavi', 'Age': '20', 'Branch': 'BE-CSE'}
{'Name': 'Dhanushka', 'Age': '21', 'Branch': 'ECE'}

===== Student Record Management =====
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter Choice: 1
Enter Name: Vaishnavi
Enter Age: 5
Enter Branch: cse
Student Added Successfully.

===== Student Record Management =====
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter Choice: 4
Enter Name to Delete: Vaishnavi
Student Deleted Successfully.

===== Student Record Management =====
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter Choice: 5
Thank You!
PS C:\Users\HP\Desktop\WEEK_01 ASSIGNMENT TASKS> |
```

CONCLUSION

This assignment helped me understand the fundamental concepts of Python programming, including variables, data types, operators, conditional statements, loops, functions, strings, collections, and file handling. Developing the Student Record Management System improved my logical thinking and programming skills by applying these concepts to solve a real-world problem. Overall, this assignment strengthened my foundation in Python and prepared me for learning Data Analytics and more advanced programming concepts.

Google Drive Folder Link:

<https://drive.google.com/drive/folders/1B-Podau3fwBVokBl8A1DDah6xsQHUiKl?usp=sharing>